Entropy Walls and Sphaleron Windows: Baryogenesis in Modulus-Dominated Universes and the Large Volume Scenario

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Abstract

Light moduli generically dominate the early universe and release entropy when they decay. We ask which baryogenesis mechanisms survive, and apply the answer to the Large Volume Scenario (LVS), assuming a single dominant modulus, no thermal inflation, and the explicit AdS vacua with $W_0 = 1$ described in the text. (i) An asymmetry produced before the decay is suppressed by the entropy released, and electroweak sphalerons are in equilibrium only in a finite temperature window, which we compute with the lattice rate; the window closes for reheating temperatures below 2.6 MeV, baryon-violating decays survive intact for $T_R \lesssim 30$ GeV, and lepton-violating decays are converted with an efficiency that scales as T_R^4. (ii) Any axion-driven, sphaleron-mediated mechanism requires an anomalous decay constant $F \lesssim 4 \times 10^{10}$ GeV, which excludes the bulk and fibre axions of LVS ($F \sim 10^{16}$ GeV). In explicit Swiss-cheese and K3-fibred vacua, electroweak baryogenesis before the decay requires $\mathcal{V} \lesssim 4 \times 10^7$ and 7×10^6, which excludes TeV-scale gravitino masses. (iii) What survives is baryon-violating modulus decay with 10^{-4}–10^{-3} baryons per decay, as found for IIB, heterotic and G_2 compactifications by Kane and Winkler. In LVS the gaugino channel that supplies it there is closed, and the higgsino channel is suppressed unless the higgsino mass is tuned. Affleck–Dine baryogenesis from a baryonic flat direction reaches the observed asymmetry for $\mathcal{V} \lesssim 6 \times 10^5$ only if the flat direction is displaced beyond the string scale; within the effective theory, $\mathcal{V} \lesssim 10^4$.

1 Introduction

String compactifications contain moduli that are light compared with the string scale and couple to matter only through Planck-suppressed interactions. Their coherent oscillations come to dominate the energy density before decay: the cosmological moduli problem [1, 2, 3]. In the Large Volume Scenario [4, 5] the volume modulus has mass $m \sim M_{\rm Pl}\mathcal{V}^{-3/2}$ and the gravitino mass is $m_{3/2} \sim M_{\rm Pl}/\mathcal{V}$. TeV-scale soft terms need $\mathcal{V} \sim 10^{14}$–10^{15}, for which m is of order 10–100 keV and the modulus decays after BBN unless it is diluted [6].

Three facts constrain baryogenesis in a modulus-dominated universe. We call the suppression of any pre-decay asymmetry by the entropy released in the decay the *entropy wall*, and the range of temperatures in which electroweak sphalerons are in equilibrium the *sphaleron window*. First, an asymmetry produced before the decay is diluted by the entropy released. Second, the Hubble rate during the EMDE is far larger than in radiation domination at the same temperature, so electroweak sphalerons may or may not be in equilibrium when one expects. Third, an asymmetry sourced by a CP-odd background is limited by how fast that background can move. The aim of this paper is to make each statement quantitative and to follow the consequences for LVS.

Results. (1) The sphaleron-equilibrium window in an EMDE is computed with the lattice rate; it closes for $T_R \lesssim 2.6$ MeV (Sec. 3). (2) Baryon-violating decays are converted with efficiency $\eta_B \simeq 1$ for $T_R \lesssim 30$ GeV, while lepton-violating decays are converted with $\eta_L \propto T_R^4$ (Sec. 3.3). (3) Any axion-driven, sphaleron-mediated mechanism obeys $F \lesssim 4.2 \times 10^{10}$ GeV (Sec. 4). (4) For asymmetries generated at temperature $T_{\rm gen}$ before the decay, $T_R \gtrsim 6\,{\rm GeV}\,(T_{\rm gen}/131.7\,{\rm GeV})$ (Sec. 5). (5) These requirements are translated into bounds on $\mathcal{V}$ in explicit Swiss-cheese and K3-fibred vacua, and, for sequestered soft terms, into the CP asymmetry that baryon-violating modulus decays must supply and its comparison with the known upper range of $\epsilon_{\rm CP}$ (Sec. 6). (6) The Affleck–Dine mechanism is analysed with attention to the validity of the effective field theory (Sec. 7).

Relation to earlier work. Kane and Winkler [22] showed that modulus decay erases any earlier asymmetry, that electroweak baryogenesis and leptogenesis then fail in the IIB, heterotic and G_2 constructions they consider, and that 10^{-4}–10^{-3} baryons per modulus decay are required; they supply them through the chain modulus $\to$ gaugino pair $\to$ *udd*. The dilution $Y_B \propto (T_R/T_c)^5$ and the existence of a maximal sphaleron temperature are due to [10, 12]; the suppression of lepton asymmetries at low reheating temperature to [11]; the relation between axion rotation, PQ charge and baryon number to [14, 15, 16]; Affleck–Dine baryogenesis in the presence of late-decaying moduli to [23, 19, 24], and in LVS to [20]; baryogenesis from gaugino decay in a thermal universe to [25, 26]; baryogenesis in an EMDE with sphaleron washout to [27]. What we add is: the sphaleron window and the efficiencies $\eta_{B,L}$ in closed form with the lattice rate; the multi-axion kinetic bound and its dark-matter version; explicit LVS vacua and the volume bounds channel by channel; the explicit supergravity result that the gaugino channel is closed in LVS and the computation of the Higgs-sector couplings; the check of the Affleck–Dine analysis against the validity of the effective theory; and an explicit amplitude calculation of the CP asymmetry, which differs from [22] in two respects (Appendix D).

Throughout, $M_{\rm Pl} = 2.435 \times 10^{18}$ GeV is the reduced Planck mass and $Y_B^{\rm obs} = 8.6 \times 10^{-11}$.

2 Moduli domination and entropy production

Let the lightest long-lived modulus ϕ, of mass m, decay with width Γ to radiation:

$$\dot\rho_\phi + 3H\rho_\phi = -\Gamma\rho_\phi, \qquad \dot\rho_R + 4H\rho_R = \Gamma\rho_\phi, \qquad H^2 = \frac{\rho_\phi + \rho_R}{3M_{\rm Pl}^2}. \tag{1}$$

We define T_R by $\Gamma = H_{\rm RD}(T_R)$,

$$\Gamma = \sqrt{\frac{\pi^2 g_R}{90}}\,\frac{T_R^2}{M_{\rm Pl}}, \qquad g_R \equiv g_*(T_R). \tag{2}$$

We assume that ϕ dominates before it decays, as is generic for an initial displacement of order $M_{\rm Pl}$; for a small displacement the dilution is absent but BBN and dark radiation constrain the model, and our bounds are for the dominating case.

For $H \gg \Gamma$ the radiation reaches a quasi-stationary state with $\rho_R \propto a^{-3/2}$. Then $(4-\frac{3}{2})H\rho_R = \Gamma\rho_\phi$, so $\rho_R = \frac{6}{5}\Gamma M_{\rm Pl}^2 H$ and

$$H(T) = \frac{\pi^2 g_*(T)}{36}\frac{T^4}{\Gamma M_{\rm Pl}^2} = \kappa\,\frac{T^4}{T_R^2 M_{\rm Pl}}, \qquad \kappa = \frac{\pi^2 g_*(T)/36}{\sqrt{\pi^2 g_R/90}}, \qquad T \propto a^{-3/8}. \tag{3}$$

For $g_* = 106.75$, $g_R = 10.75$ the coefficient exceeds the usual $\sqrt{\pi^2 g_*/90}$ by a factor 7.9. We have checked Eq. (3) against numerical solutions of Eq. (1) at $T = 131.7$ GeV to better than 0.1%.

Dilution. The comoving entropy $S = sa^3$ scales as T^{-5} in this regime. An asymmetry frozen into the comoving volume at radiation temperature $T_{\rm gen}$ is reduced by

$$\mathcal{D} \equiv \frac{S(T_{\rm gen})}{S_{\rm fin}} = \mathcal{C}\left(\frac{T_R}{T_{\rm gen}}\right)^5, \qquad \mathcal{C} \simeq 0.16\,\frac{g_R}{g_*(T_{\rm gen})}. \tag{4}$$

The coefficient follows from matching the sudden-decay entropy to Eq. (3) (Appendix A); it gives 0.016 for $T_R \lesssim 50$ MeV and 0.11–0.14 for $0.2 \lesssim T_R \lesssim 60$ GeV, in agreement with the numerical solution (Fig. 1). For $T_R = 18$ MeV, $\mathcal{D} = 7.6 \times 10^{-22}$.

Validity: primordial radiation. Equation (3) assumes that the radiation at $T_{\rm gen}$ is produced by the decays. If the modulus starts to oscillate at $H = m$ in a bath of primordial radiation left by the inflaton, that radiation redshifts as a^{-1} in temperature and dominates the temperature until it falls below the decay product, which cools as $a^{-3/8}$. With $\rho_\phi = \rho/6$ at $H = m$, the crossover occurs at

$$T_\times \simeq 5.4 \times 10^2\ {\rm GeV}\left(\frac{m}{10^7\ {\rm GeV}}\right)^{13/10}. \tag{5}$$

The steady-state description at $T_{\rm gen} = 131.7$ GeV requires $T_\times \gtrsim T_{\rm gen}$, that is $m \gtrsim 3.4 \times 10^6$ GeV or $T_R \gtrsim 0.7$ GeV. All bounds quoted below that rely on $T_R \gtrsim 6$ GeV satisfy this. For smaller T_R the numbers in Table 1 and Fig. 2 depend on the pre-oscillation history; the qualitative conclusions do not, since in that case the true $H(T_*)$ is smaller and $\Gamma/H(T_*) \ll 1$ remains.

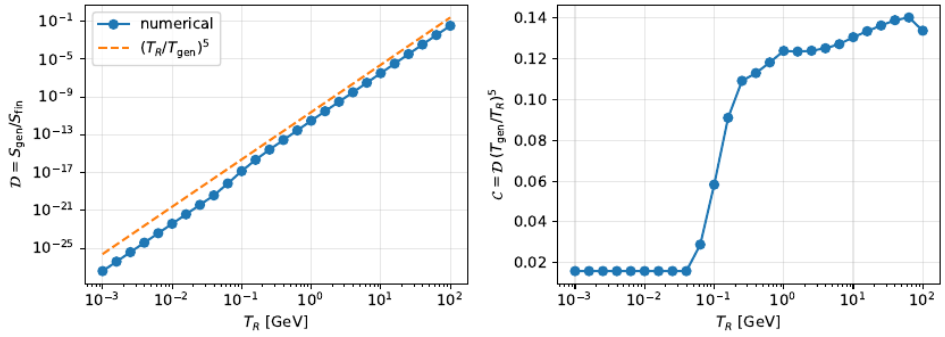


Figure 1: Entropy dilution $\mathcal{D} = S(T_{\rm gen})/S_{\rm fin}$ for $T_{\rm gen} = 131.7$ GeV from Eq. (1) (left) and the prefactor $\mathcal{C}$ (right). The step near $T_R \sim 0.1$ GeV is the change of g_* across the QCD crossover in our approximate $g_*(T)$.

3 Electroweak sphalerons in an EMDE

3.1 Susceptibility

A sphaleron transition changes Chern–Simons number by $\Delta N_{\rm CS} = \pm 1$ and produces twelve left-handed doublet fermions. A bias $\mu_{\rm CS}$ conjugate to $N_{\rm CS}$ enforces $n_g(3\mu_Q + \mu_L) = \mu_{\rm CS}$. Imposing Yukawa equilibrium ($\mu_u = \mu_Q + \mu_H$, $\mu_d = \mu_Q - \mu_H$, $\mu_e = \mu_L - \mu_H$), hypercharge neutrality and $B - L = 0$, one finds (Appendix B)

$$n_B = \tfrac{12}{79}T^2\mu_{\rm CS} \quad \Rightarrow \quad Y_B = \xi\,\frac{\mu_{\rm CS}}{T}, \qquad \xi = \frac{12/79}{2\pi^2 g_*/45} = 3.24 \times 10^{-3}. \tag{6}$$

With $\mu_{\rm CS} = 0$ the same system gives the standard $B = \frac{28}{79}(B-L)$. In an EMDE the Hubble rate at $T \sim 150$ GeV is enhanced by $7.9\,T^2/T_R^2$ relative to radiation domination, and the electron Yukawa coupling is out of equilibrium for $T_R \lesssim 15$ GeV. Removing e_R from the equilibrium network lowers ξ by 4%.

3.2 Rate and equilibrium window

The lattice rate for $m_H = 125$ GeV is [18]

$$\frac{\Gamma_{\rm sph}}{T^4} = \begin{cases} \exp(0.83\,T/\text{GeV} - 147.7), & 130\ \text{GeV} < T < T_c, \\ 18\,\alpha_W^5, & T > T_c, \end{cases} \qquad T_c = 159\ \text{GeV}. \tag{7}$$

In radiation domination sphalerons freeze out at $T_* = 131.7 \pm 2.3$ GeV, where the washout rate $\lambda_{\rm sph} = c^{-1}\Gamma_{\rm sph}/T^3$ equals H. We fix $c = 0.12$ (i.e. $1/c \simeq 8.5$) to reproduce this value; it is a calibration, not a prediction, and we vary it by a factor of ten below. In an EMDE H is given by Eq. (3) and grows as T^4, whereas $\Gamma_{\rm sph}/T^3$ grows only as T above T_c. Equilibrium therefore holds only in a window

$$T_*(T_R) < T < T_{\rm up}(T_R), \qquad T_{\rm up}^3 = \frac{18\,\alpha_W^5\,T_R^2 M_{\rm Pl}}{c\,\kappa}. \tag{8}$$

The scaling $T_{\rm up} \propto T_R^{2/3}$ is due to [10]; the lattice rate fixes the coefficient and the broken-phase fit fixes the lower edge (Table 1). The window closes when $T_{\rm up} = T_c$, i.e.

$$T_R^{\rm eq} = \left(\frac{c\,\kappa\,T_c^3}{18\,\alpha_W^5 M_{\rm Pl}}\right)^{1/2} = 2.6\ \text{MeV} \tag{9}$$

(0.8–8.1 MeV for $c \to c/10,\ 10c$), at or below the BBN bound. The fit (7) is discontinuous at T_c by a factor ~ 5, so edges near T_c are uncertain by a few GeV.

T_R	18 MeV	56 MeV	1 GeV	10 GeV
T_* [GeV]	156	153	145	139
$T_{\rm up}$ [GeV]	582	1296	$\gtrsim 3000$	$\gtrsim 3000$

Table 1: Sphaleron equilibrium window in an EMDE.

3.3 Washout and conversion efficiencies

Let each modulus decay create baryon number ϵ_B and lepton number ϵ_L. With comoving densities $N_X = n_X a^3$, $N_\phi = \rho_\phi a^3/m$,

$$\dot N_{B-L} = (\epsilon_B - \epsilon_L)\Gamma N_\phi, \qquad \dot N_B = \epsilon_B \Gamma N_\phi - \lambda_{\rm sph}(T)\Big(N_B - \tfrac{28}{79} N_{B-L}\Big), \tag{10}$$

to be solved with Eq. (1). We use the symmetric-phase target $\frac{28}{79}$; the broken-phase correction near freeze-out is a few percent. The system is linear in the sources, so $N_B^{\rm fin} = \eta_B \epsilon_B N_\phi^{\rm tot}$ for $\epsilon_L = 0$ and $N_B^{\rm fin} = -\eta_L \epsilon_L N_\phi^{\rm tot}$ for $\epsilon_B = 0$, with $\eta_{B,L}$ depending only on T_R (Fig. 2).

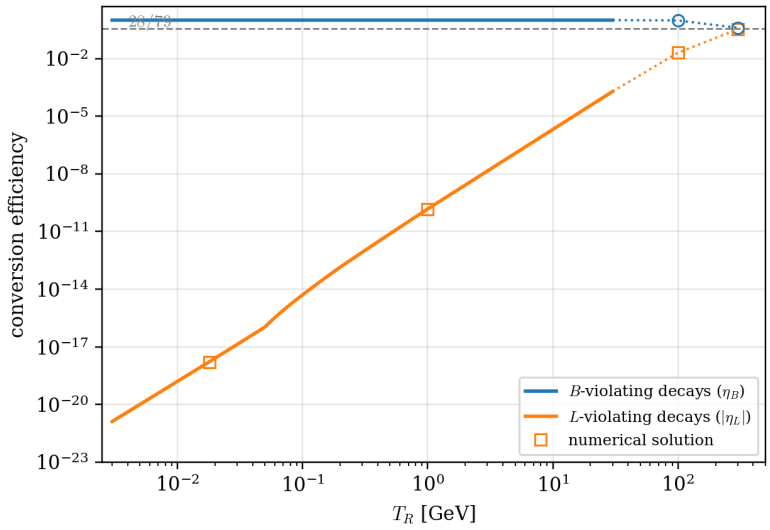


Figure 2: Conversion efficiencies. Curves: closed forms (11) for $T_R \leq 30$ GeV; open markers: numerical solutions of Eq. (10); dotted lines join the numerical values at larger T_R. Baryon-violating decays survive for $T_R \lesssim 30$ GeV; lepton asymmetries are converted only if an appreciable fraction of the decays occurs while sphalerons are in equilibrium.

Closed forms. For $T_R \ll T_*$ the fraction of the modulus energy that has decayed by $t = 2/3H$ is $\Gamma t = \frac{2}{3}\Gamma/H$. Lepton number produced before sphaleron freeze-out is converted with coefficient $28/79$, and all later decays leave B untouched. Using Eqs. (2) and (3), $\Gamma/H = \frac{2g_R}{5g_*}(T_R/T)^4$, so

$$\eta_L \simeq \frac{28}{79}\cdot\frac{2}{3}\frac{\Gamma}{H(T_*)} = \frac{28}{79}\,\frac{4}{15}\,\frac{g_R}{g_*}\left(\frac{T_R}{T_*}\right)^4, \qquad \eta_B \simeq 1 - \frac{51}{79}\cdot\frac{2}{3}\frac{\Gamma}{H(T_*)}. \tag{11}$$

Here $T_* = T_*(T_R)$ is the lower edge of the window from Table 1; with it, Eq. (11) gives 1.7×10^{-18}, 1.5×10^{-10}, 2.0×10^{-6} at $T_R = 18$ MeV, 1 GeV, 10 GeV, within 10% of the numerical solution. (Using the radiation-domination value $T_* = 131.7$ GeV overestimates η_L by up to a factor 2.) Numerically $\eta_B = 1$ to better than 10^{-3} for $T_R \lesssim 30$ GeV, falling to 0.96 at 100 GeV and 0.40 at 300 GeV; $\eta_L \simeq 1.4\times10^{-10}(T_R/\text{GeV})^{4.1}$ for 1–10 GeV, reaching 0.02 at 100 GeV and 0.33 at 300 GeV. Leptogenesis from modulus decay therefore needs $T_R \gtrsim 100$–300 GeV; for $T_R \lesssim 30$ GeV the sphalerons are spectators. Multiplying Eq. (11) by the decay yield $\epsilon\kappa_Y T_R/m$ gives $Y_B \propto \epsilon T_R^5/(T_*^4 m)$, the scaling found for non-thermal leptogenesis in [11].

4 Axion-driven sources

Let $\theta^i = a^i/f$ have kinetic term $\frac{1}{2}G_{ij}\dot\theta^i\dot\theta^j$ and couple through $\mathcal{L} \supset \frac{g^2}{32\pi^2}c_i\theta^i W\tilde W$ with constant c_i. In string compactifications $c_i = k_i$ are the integers in $f_W = k_i T^i$. Since $\frac{g^2}{32\pi^2}W\tilde W = \partial_\mu K^\mu$,

$$\mu_{\rm CS} = \frac{d}{dt}(c_i\theta^i) = c_i\dot\theta^i. \tag{12}$$

Remark (frame independence). $c_i\theta^i$ is a scalar, so $\mu_{\rm CS}$ is unchanged by any time-dependent rotation $R(t)$ of the field-space basis: $\tilde c_\alpha\dot{\tilde\theta}^\alpha + \dot{\tilde c}_\alpha\tilde\theta^\alpha = c_i\dot\theta^i$. In particular a rotating eigenbasis, such as an

avoided crossing in a thermal mass matrix, produces no chemical potential unless the axions acquire a coordinate velocity.

Define $F \equiv (c_i G^{ij} c_j)^{-1/2}$, with G^{ij} the inverse of G_{ij}; for one axion with $\theta = a/f$, $F = f/N$.

Bound. If sphalerons are in equilibrium at $T_{\rm gen}$ and $\rho_{\rm kin} = \frac{1}{2} G_{ij}\dot\theta^i\dot\theta^j \le \rho_{\rm tot} = 3H^2 M_{\rm Pl}^2$, the Cauchy–Schwarz inequality in the metric G gives $|c_i\dot\theta^i| \le F^{-1}\sqrt{2\rho_{\rm kin}}$, hence

$$Y_B^{\rm gen} \le \xi\,\frac{\sqrt{6}\,H(T_{\rm gen})M_{\rm Pl}}{F\,T_{\rm gen}} \xrightarrow{\rm RD} \xi\,\pi\sqrt{\frac{g_*}{15}}\frac{T_{\rm gen}}{F} \simeq \frac{3.6\ {\rm GeV}}{F}, \qquad F \le F_{\rm max} = 4.2\times 10^{10}\ {\rm GeV}. \quad (13)$$

Only the sphaleron freeze-out region contributes, because $B+L$ relaxes to its equilibrium value while sphalerons are active and $B-L$ is not generated by $\mu_{\rm CS}$; thus $T_{\rm gen} = T_*(T_R)$. The bound uses no information about the source and is saturated only if the axion kinetic energy equals the total energy density (kination). In an EMDE, $Y_B^{\rm fin} \propto H_{\rm EMDE}\mathcal{D} \propto T_R^3$, so an EMDE only weakens the mechanism: for $T_R \le 30$ GeV the maximal yield is lower by 4.9×10^{-3}, $F_{\rm max} \to 2\times 10^8$ GeV. A source limited by the Hubble rate, $|c_i\dot\theta^i| \le \nu H$, gives $Y_B \le \xi\nu H/T$, which at $T = 131.7$ GeV requires $\nu \gtrsim 10^8$.

For a single axion, with $n_\theta = f^2\dot\theta$ and $Y_\theta = n_\theta/s$, the same susceptibility gives $Y_B = \frac{12}{79}N(T^2/f^2)Y_\theta$, the relation underlying axiogenesis [14].

Dark-matter-constrained version. If the rotation ends as cold axion dark matter with conserved PQ charge, $m_a Y_\theta^{\rm fin} \le \rho_{\rm DM}/s = 4.4\times 10^{-10}$ GeV. Baryons and axions are diluted equally, so for any EMDE history

$$\frac{f^2 m_a}{N} \le \frac{(12/79)\,T_{\rm gen}^2(\rho_{\rm DM}/s)}{Y_B^{\rm obs}} = 1.35\times 10^4\ {\rm GeV}^3. \quad (14)$$

For the QCD axion, $m_a f \simeq 5.7\times 10^{-3}$ GeV^2 gives $f \lesssim 2\times 10^6 N$ GeV. This is more restrictive than the values 10^7–10^8 GeV quoted in the axiogenesis literature. The bound can be evaded by dissipating the rotation [16], which is why Eq. (13), which assumes nothing about the endpoint, is the more robust statement.

Classification. Let $H_* = H(T_*)$. (A) $m_a \lesssim H_*$: the field is frozen, $|\dot\theta| \lesssim H_*\theta$, and the Hubble-limited bound applies. (B) $m_a \gg H_*$, $\Gamma_a \ll H_*$: the axion oscillates and is long lived; Eq. (13) applies. (C) $\Gamma_a \gtrsim H_*$: the oscillation is suppressed by $e^{-\Gamma_a/4H_*}$ and no coherent velocity survives at freeze-out.

5 Generic bound on T_R for pre-decay generation

Let a mechanism act at $T_{\rm gen}$ with maximal chemical potential $(\mu/T)_{\rm max}$. Then $Y_B^{\rm fin} \le \mathcal{D}(T_R)\,\xi\,(\mu/T)_{\rm max}$, and

$$T_R \ge T_R^{\rm min} \simeq T_{\rm gen}\left[\frac{Y_B^{\rm obs}}{\mathcal{C}\,\xi\,(\mu/T)_{\rm max}}\right]^{1/5}. \quad (15)$$

For $T_{\rm gen} = 131.7$ GeV: $T_R^{\rm min} = 6.1, 9.6, 15.1, 23.9, 37.7$ GeV for $(\mu/T)_{\rm max} = 1, 10^{-1}, \ldots, 10^{-4}$. Since $T_R^{\rm min} \propto T_{\rm gen}$, mechanisms operating at lower temperature are less constrained; the bound applies to generation at the electroweak scale or above. In terms of the modulus mass, with $\Gamma = c_\Gamma m^3/4\pi M_{\rm Pl}^2$ and $c_\Gamma = 1$, $T_R \ge 6.1$ GeV is $m_\Phi \gtrsim 1.5\times 10^7$ GeV (scaling as $c_\Gamma^{-1/3}$), about two orders of magnitude above the value $m_\Phi \gtrsim 9\times 10^4$ GeV required by BBN for $T_R \ge 5$ MeV.

6 Application to the Large Volume Scenario

6.1 Explicit vacua

We use the Swiss-cheese compactification on $\mathbb{P}^4_{[1,1,1,6,9]}$ ($h^{1,1}=2$, $\chi=-540$) [4], with

$$K=-2\ln\!\big(\mathcal{V}+\tfrac{\hat{\xi}}{2}\big),\quad \mathcal{V}=\gamma(\tau_b^{3/2}-\tau_s^{3/2}),\quad W=W_0+Ae^{-aT_s}, \tag{16}$$

$\gamma=1/9\sqrt{2}$, $\hat{\xi}=-\zeta(3)\chi/2(2\pi)^3 g_s^{3/2}$, $W_0=|A|=1$, $g_s=0.3$ (0.4 for the smallest volumes), $a=2\pi/N$ treated as continuous. The exact $\mathcal{N}=1$ potential is minimised at high precision and the Hessian checked to be positive definite. The vacua are AdS, which changes masses at $O(1)$ relative to a de Sitter vacuum. Ten vacua cover $\mathcal{V}=2.5\times10^3$–$10^{16}$. They confirm $m_{3/2}\simeq M_{\rm Pl}/\mathcal{V}$, $m_{\tau_b}=c_m M_{\rm Pl}\mathcal{V}^{-3/2}$ with $c_m=1.0$–2.1, and $m_{\tau_s}\simeq m_\rho\simeq(17\text{–}78)\,m_{3/2}$; a modulus mass of 10^5 GeV corresponds to $\mathcal{V}\sim10^9$, not 10^{15}. We take $\Gamma=c_\Gamma m^3/4\pi M_{\rm Pl}^2$ with $c_\Gamma=1$ (varied between 0.3 and 8). The width to the bulk axion partner, $m^3/48\pi M_{\rm Pl}^2$ [7], produces dark radiation; $\Delta N_{\rm eff}<0.3$ requires $c_\Gamma\gtrsim0.9$–1.8 [8].

For the K3-fibred model [9] the fibre modulus is the lightest. With the string-loop potential of [9], which is conjectural, we find $m_f=c_f M_{\rm Pl}\mathcal{V}^{-5/3}$ with $c_f=1.45$ for the reference loop data (0.70–3.0 under factor-of-three variations; 17 for the published TeV-scale parameter set), so $T_R\propto M_{\rm Pl}\mathcal{V}^{-5/2}$. For the published LV set the numerical fibre mass reproduces the analytic $m_f^2=4(1+\frac{7}{6}R)C_2^2/\mathcal{V}^{10/3}$ to fourteen digits. If heavier moduli decay earlier, additional dilution can only reduce the final asymmetry, so the bounds below hold for multi-stage histories.

Maximal $\mathcal{V}$ for	$T_R\geq6.1$ GeV	$T_R\geq100$ GeV	$T_R\geq5$ MeV
Swiss-cheese	3.8×10^7	1.1×10^7	1.1×10^9
K3-fibred, $c_f=1.45$	6.6×10^6	2.1×10^6	1.4×10^8
K3-fibred, $c_f=0.7$–3.0	4.3×10^6–1.0×10^7	1.4×10^6–3.3×10^6	9.0×10^7–2.2×10^8

Table 2: Upper bounds on $\mathcal{V}$ from pre-decay electroweak baryogenesis, decay leptogenesis, and BBN.

6.2 Channel by channel

Pre-decay electroweak baryogenesis requires $T_R\gtrsim6$–38 GeV, i.e. $\mathcal{V}\lesssim3.8\times10^7$ (Swiss-cheese) and 7×10^6 (K3), $m_{3/2}\gtrsim10^{10}$ GeV. At $\mathcal{V}=10^{16}$, $T_R\simeq6\times10^{-19}$ GeV.

Axion-driven baryogenesis. For $f_W=NT$ with T a bulk or fibre modulus, $K_{T\bar{T}}=O(1)/\tau^2$ and $\alpha_W^{-1}=N\tau$, so

$$F=\frac{\kappa_F\,\alpha_W M_{\rm Pl}}{4\pi},\qquad \kappa_F=2\tau\sqrt{2K_{T\bar{T}}}=O(1)\ (\sqrt{2}\text{ for the fibre}), \tag{17}$$

independent of N and τ: $F\simeq9.3\times10^{15}$ GeV, 2×10^5 times the bound (13), even at kination. For the Standard Model on a blow-up cycle, $K_{s\bar{s}}\simeq3\gamma/8\sqrt{\tau_s}\mathcal{V}$ and $F_s\simeq4.0\times10^{16}\,{\rm GeV}/\sqrt{\mathcal{V}}$, which satisfies (13) for $\mathcal{V}\gtrsim9\times10^{11}$. But the axion is heavy, $m_\rho\simeq(17\text{–}78)m_{3/2}$, and decays to gauge bosons with $\Gamma_\rho\simeq\frac{3}{8}\alpha_W^2 m_\rho^3/32\pi^3F_s^2$; in the explicit vacua $\Gamma_\rho/H(T_*)\geq2.3\times10^3$ for all $\mathcal{V}$, so it is of type (C) and provides no velocity. (For $\mathcal{V}\gtrsim10^{10}$ the radiation at T_* is not in the steady state, but there T_R is far below BBN in any case.)

Lepton-violating decays. $\eta_L\simeq1.4\times10^{-10}(T_R/{\rm GeV})^{4.1}$, so $T_R\gtrsim100$ GeV is needed: $\mathcal{V}\lesssim1.1\times10^7$.

Baryon-violating decays. With ϵ the CP-violating baryon number per decay, $Y_B = \epsilon\,\eta_B \kappa_Y T_R/m$ with $\kappa_Y \simeq 0.7$–0.9. Since $T_R/m \propto \mathcal{V}^{-3/4}$, $\mathcal{V}_{\max} \propto \epsilon^{4/3}$; in the explicit vacua $\mathcal{V}_{\max} = 3.1\times10^{12}$, 7.3×10^{9}, 2.6×10^{6}, 6.5×10^{3} for $\epsilon = 1$, 10^{-2}, 10^{-4}, 10^{-6}. TeV-scale LVS requires $\epsilon \gg 1$. The value of ϵ is model dependent; mechanisms in which it is fixed, through the decay chain modulus $\to$ gaugino pair $\to$ *udd*, are described in [21, 22] for KKLT-type and other stabilisation schemes; the LVS case is considered next.

6.3 Sequestered soft terms and the required CP asymmetry

In sequestered LVS the soft terms on the Standard Model brane are suppressed relative to $m_{3/2} \sim M_{\rm Pl}/\mathcal{V}$ by at least $\mathcal{V}^{-1/2}$ and may be as small as $M_{\rm Pl}/\mathcal{V}^2$; gaugino masses are of order $m_{3/2}/\mathcal{V}$, and scalar masses are either comparably small or larger by $\mathcal{V}^{1/2}$ [17]. For TeV soft terms [17] quotes $\mathcal{V} \sim 10^6$–10^7; the precise value depends on W_0 and g_s. For the vacua used here ($W_0 = 1$), $m_{\rm soft} = M_{\rm Pl}/\mathcal{V}^2$ lies between 10 TeV and 0.1 TeV for $1.5\times10^7 \lesssim \mathcal{V} \lesssim 1.5\times10^8$. In this range $m_{3/2} \sim 10^{10}$–10^{11} GeV, $m_{\tau_b} \simeq 2\times10^6$–$6\times10^7$ GeV and $T_R \simeq 0.4$–50 GeV, so that $\eta_B \simeq 1$ (Sec. 3.3), and decays to superpartners are kinematically open. Solving $Y_B = \epsilon\,\eta_B\kappa_Y T_R/m$ for ϵ gives (Fig. 3)

$$\epsilon \simeq 1.3\times10^{-4},\ 2.8\times10^{-4},\ 5.6\times10^{-4} \qquad \text{at } \mathcal{V} = 1.5\times10^7,\ 5\times10^7,\ 1.5\times10^8, \tag{18}$$

with a minimum $\epsilon \simeq 2\times10^{-4}$ near $\mathcal{V} = 3\times10^7$.

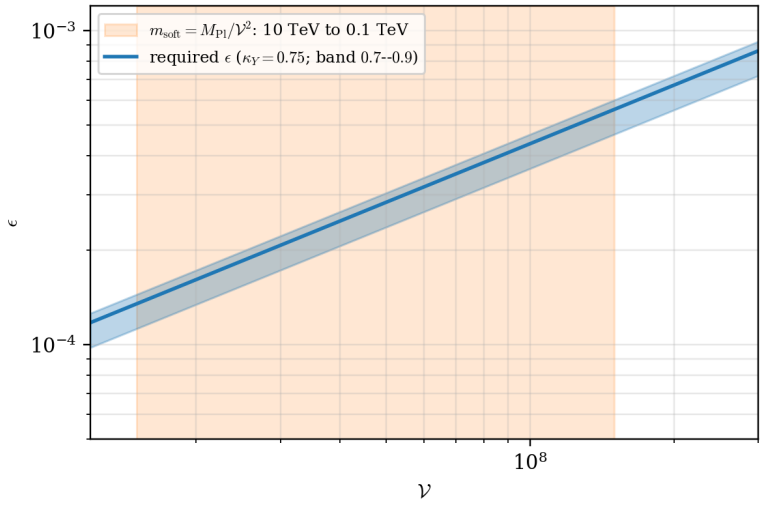


Figure 3: CP-violating baryon number per decay required to reproduce $Y_B^{\rm obs}$ from baryon-violating decays of the volume modulus in the explicit Swiss-cheese vacua, for $T_R \lesssim 100$ GeV where $\eta_B \simeq 1$.

Counting and phases. If N_f fermions of the decaying species are produced per modulus decay into the pair, and each decays to tsb or $\bar{t}\bar{s}\bar{b}$ with CP asymmetry $\epsilon_{\rm CP}$, then $\epsilon = N_f\,{\rm Br}\,\epsilon_{\rm CP}$, with Br the branching ratio of the modulus to the pair. For gluinos $N_f = \frac{4}{3}$ (two gluinos per pair, eight of the twelve gauginos); for higgsinos $N_f = 2$. We write $\phi_{12} = \arg(m_{\lambda_2} m^*_{\lambda_1})$ for the relative phase of the two Majorana masses; the quantity written $\sin 2\phi_{12}$ in [22] is $\sin\phi_{12}$ here, as follows from matching their f_2 term (Appendix D).

Gauge bosons and gauginos. The widths follow from the supergravity data of the vacua, with the formulas of [22], the light eigenstate Φ of $G^{-1}\partial^2 V$ being normalised canonically and $\delta\tau^i = v^i\phi$. The width to the massless axion of T_b is $\Gamma_{aa} = g^2 m^3/128\pi M_{\rm Pl}^2$ with $g = \partial_\phi K_{b\bar b}/K_{b\bar b}$ and $g^2 = \frac{8}{3}$, that is $m^3/48\pi M_{\rm Pl}^2$; we confirm this numerically to four digits in each vacuum. If the gauge kinetic function is $f = T_s$, as for a stack wrapping the blow-up cycle, then $\Gamma_{gg}/\Gamma_{aa} = 0.03$–$0.06$ (Table 3) and the gaugino mass $M_{\tilde\lambda} = F^{T_s}/2\tau_s$ exceeds m_Φ by a factor 17–630, which grows as $\mathcal{V}^{1/2}$; the gaugino-pair channel is closed. If f does not depend on T_b or T_s, as for D3-branes at a singularity at leading order, then $\partial_\Phi f = 0$ and Γ_{gg} and $\Gamma_{\tilde\lambda\tilde\lambda}$ vanish at tree level; the corresponding couplings arise only at loop level and from anomaly-mediated terms, which are further suppressed. In neither case is $\mathrm{Br}_{\tilde\lambda\tilde\lambda}$ of order one, so the gluino chain of [22] does not operate in LVS.

$\mathcal{V}$	m_Φ [GeV]	$M_{\tilde\lambda}/m_\Phi$ $(f = T_s)$	Γ_{gg}/Γ_{aa} $(f = T_s)$	$\Gamma_{\tilde h\tilde h}/\Gamma_{aa}$ $[z_0^2]$
2.6×10^5	3.0×10^{10}	17	0.063	1.6×10^{-3}
3.4×10^6	6.0×10^8	54	0.045	1.5×10^{-4}
1.2×10^7	8.3×10^7	99	0.039	4.4×10^{-5}
4.5×10^7	1.1×10^7	183	0.034	1.3×10^{-5}
1.7×10^8	1.6×10^6	339	0.029	3.8×10^{-6}
6.1×10^8	2.1×10^5	630	0.026	$\simeq10^{-6}$

Table 3: Light-modulus decay widths in the explicit Swiss-cheese vacua ($W_0 = A = 1$, $g_s = 0.3$, no uplift), from the formulas of [22] with $N_g = 12$. The last column is for $Z = z_0\hat Z$, $\hat Z \propto \mathcal{V}^{-2/3}$ (the value at 6.1×10^8 is extrapolated).

Higgs sector. In [22] the widths to higgsinos and Higgs bosons are $\Gamma_{\tilde h\tilde h} = m_\Phi|\partial_\phi\mu|^2/4\pi$ and $\Gamma_{HH} = (|\partial_\phi m_H^2|^2 + |\partial_\phi B\mu|^2)/4\pi m_\Phi$, where $\mu = (m_{3/2}Z - \bar F^{\bar m}\partial_{\bar m}Z)/\hat Z$ for a Giudice–Masiero coefficient Z and matter metric $\hat Z$, and $\partial_\phi = v^k\partial_{\tau_k}$. The simplified form $\Gamma \propto m_\Phi^3|\partial Z|^2/\hat Z^2$ given there neglects terms of relative order $m_{3/2}/m_\Phi$, and does not apply to LVS, where $m_{3/2}/m_\Phi \simeq \mathcal{V}^{1/2}$. For $\hat Z \propto \mathcal{V}^{-2/3}$ and $Z = z_0\hat Z$ we have evaluated these expressions with the exact F-terms of the vacua. We find

$$\mu = -(4.0\pm0.1)\,z_0\,\frac{m_{3/2}}{\mathcal{V}} \qquad (3\times10^5 \lesssim \mathcal{V} \lesssim 2\times10^8)\,: \tag{19}$$

the leading terms of $m_{3/2}Z$ and $\bar F\partial Z$ cancel, because $F^{T_b} = -2\tau_b m_{3/2}$ and $\partial_{T_b}\ln\hat Z = -1/2\tau_b$, and μ is of TeV size for $\mathcal{V}\sim10^7$–10^8 with $z_0 = O(1)$. The cancellation holds as a function of τ_b, and so the modulus coupling is suppressed as well: $\Gamma_{\tilde h\tilde h}/\Gamma_{aa} = (4.4\times10^{-5}, 1.3\times10^{-5}, 3.8\times10^{-6})\,z_0^2$ at $\mathcal{V} = 1.2\times10^7, 4.5\times10^7, 1.7\times10^8$. Thus $\mathrm{Br}_{\tilde h\tilde h} \lesssim 10^{-4}z_0^2$, and $z_0 \gtrsim 10^2$ would be needed for $\mathrm{Br}_{\tilde h\tilde h} \simeq 0.4$. The Higgs soft masses for this matter metric are $|m_H| \simeq 3.2\,m_{3/2}/\sqrt{\mathcal{V}}$, far above the TeV scale, so a TeV-scale Higgs sector requires additional sequestering, whose effect on these couplings we do not determine. If instead μ is made TeV-scale by a tuned cancellation, $\partial_\phi\mu \sim m_{3/2}$, $\Gamma_{\tilde h\tilde h} \gg \Gamma_{aa}$, and $\mathrm{Br}_{\tilde h\tilde h} \simeq 1$, at the price of a tuning of μ at the level $\mathcal{V}^{-1}$.

Required asymmetry. For higgsinos $\epsilon_{\rm CP} = \epsilon/2\mathrm{Br}_{\tilde h\tilde h}$: (0.65–2.8) $\times10^{-4}$ for $\mathrm{Br}_{\tilde h\tilde h} = 1$, (1.6–7) $\times10^{-4}$ for 0.4, and more than 10 for 10^{-5}, which is impossible.

The CP asymmetry. We have computed it by an explicit amplitude calculation (Appendix D). The neutral higgsino couples to the top and the stop through $W \supset y_t H_u Q U^c$, with the structures $(\tilde h\psi_Q)\tilde t_R^*$ and $(\tilde h\psi_{U^c})\tilde t_L$, each with coefficient $y_t/\sqrt{2}$ for a Majorana mass eigenstate; unlike the

gluino, it couples the top of one chirality to the stop of the other. With the stops integrated out, the amplitudes are governed by the propagator matrix $D = (M_{\tilde{t}}^2)^{-1}$ in the $(\tilde{t}_R, \tilde{t}_L)$ basis. The absorptive part comes from the cut through $\tilde{g}\,\bar{t}$, and the CP-odd term is proportional to $m_{\tilde{h}} m_{\tilde{g}} f_2(m_{\tilde{g}}^2/m_{\tilde{h}}^2) \sin\phi_{12}$. We find that the two coupling structures contribute to this term with equal magnitude and opposite sign: a gaugino mass insertion enters with the phase $e^{i\phi}$ for a coupling to $\lambda\psi$ and $e^{-i\phi}$ for a coupling to $\lambda^\dagger\psi^\dagger$. For a pure higgsino they cancel, for any stop masses, any stop mixing and any phase of X_t ($|\epsilon_{\rm CP}| < 10^{-18}$ over a scan); the analytic reason is that the two chains carry complex-conjugate phases $e^{\mp i\Phi}$, with $\Phi = \phi_{12} + 2\arg X_t$, and equal kinematic factors (Appendix D). A bino admixture a_B breaks the symmetry between the structures and gives

$$\epsilon_{\rm CP} \simeq 7\times10^{-4}\, a_B \qquad (m_{\tilde{t}} = 2\mu,\ X_t = 2.5\, m_{\tilde{t}},\ m_{\tilde{g}} = 0.4\mu,\ \phi_{12} = \pi/2,\ X_t \text{ real},\ \lambda'' \to \infty), \quad (20)$$

which reaches the required value for $a_B \gtrsim 0.1$–0.4 at $\mathrm{Br}_{\tilde{h}\tilde{h}} = 1$ (Fig. 4); for $m_{\tilde{t}} = 3\mu$ it requires $a_B \gtrsim 0.3$–0.7. For a mixed neutralino the asymmetry depends on several phases; with ϕ_{12} and $\arg X_t$ free it is larger than this benchmark by up to a factor 1.9 (at $a_B = 0.4$), and the required a_B is correspondingly smaller, $\gtrsim 0.05$–0.2. At $\lambda''_{323} = 2$, the value used in [22], these asymmetries are reduced to 73–91% of the $\lambda'' \to \infty$ values (the range reflects the normalisation of the three-body rate), and at $\lambda''_{323} = 1$ to 40–73%. If the left-handed term instead entered with the same sign as the right-handed one, as in Eq. (41) of [22], the two structures would add and a pure higgsino would give $|\epsilon_{\rm CP}| \le 1.4\times10^{-4}$, 2.7×10^{-5} and 3.5×10^{-6} at $m_{\tilde{t}}/\mu = 2$, 3, 5, falling as $m_{\tilde{t}}^{-6}$; this reaches the required value only for $m_{\tilde{t}} \lesssim 2\mu$ and $\mathrm{Br}_{\tilde{h}\tilde{h}} \simeq 1$. In either case the higgsino route needs a tuned μ with $\mathrm{Br}_{\tilde{h}\tilde{h}} \simeq 1$, together with a bino admixture or light, strongly mixed stops, and $\mu > m_{\tilde{g}}$; it is a corner of parameter space.

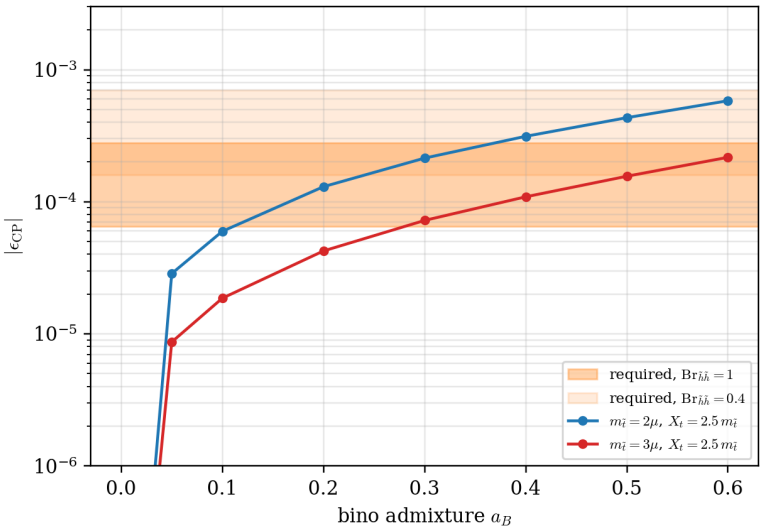


Figure 4: CP asymmetry in the decay of a neutralino, a mixture of the neutral higgsino and the bino with bino fraction a_B, to tsb through a gluino loop ($\phi_{12} = \pi/2$, X_t real, $m_{\tilde{g}} = 0.4\mu$, $\lambda'' \to \infty$). The asymmetry vanishes for $a_B = 0$. The bands are the asymmetry required in sequestered LVS for $\mathrm{Br}_{\tilde{h}\tilde{h}} = 1$ (dark) and 0.4 (light).

6.4 Other moduli stabilisation schemes

The estimate of the baryon number per decay required in the IIB, heterotic and G_2 constructions is due to [22]. The entropy wall does not depend on the construction, only on m_Φ and the total

width, and Table 4 recomputes the estimate with the formulas of this paper, extends it to LVS, and adds the dilution and conversion factors. Table 4 applies Eqs. (4) and (11) to the three classes of compactification studied in [22], using the decay coefficients $\Gamma = c\,m^3/M_{\rm Pl}^2$ listed there (IIB/KKLT: $c_{\rm tot} \simeq 0.13$, $m_\Phi = 10^6$–8×10^6 GeV; heterotic: $c_{\rm tot} \simeq 0.21$, $m_\Phi = 2$–6×10^6 GeV; G_2: $0.1 \le c_{\rm tot} \le 36$, $m_\Phi = 6\times10^4$–1.6×10^5 GeV), together with the sequestered LVS window of Sec. 6.3.

	T_R [GeV]	$\mathcal{D}(131.7\text{ GeV})$	η_L	$\epsilon_{\rm req}$	gaugino pairs
IIB / KKLT	0.16–3.0	10^{-16}–10^{-9}	10^{-14}–10^{-8}	$(3\text{–}7)\times10^{-4}$	open
heterotic	0.5–2.5	10^{-13}–10^{-10}	10^{-11}–10^{-8}	$(2.7\text{–}4.5)\times10^{-4}$	open
G_2	0.003–0.17	10^{-26}–10^{-16}	10^{-21}–10^{-13}	$(1\text{–}24)\times10^{-4}$	open
sequestered LVS	0.4–54	10^{-14}–10^{-3}	10^{-10}–0.02	$(1.3\text{–}5.6)\times10^{-4}$	closed

Table 4: Entropy-wall quantities in four moduli-dominated cosmologies. $\epsilon_{\rm req} = Y_B^{\rm obs}/Y_\Phi$ with $Y_\Phi = \frac{3}{4}T_R/m_\Phi$ is the baryon number per modulus decay required of baryon-violating decays. The last column states whether the modulus can decay to a gaugino pair: open in [22] for the first three, closed in LVS (Sec. 6.3).

In every case T_R lies below the 6.1 GeV required for electroweak-scale generation before the decay, which in these schemes translates into $m_\Phi \gtrsim 1.1$–1.4×10^7 GeV for $c_{\rm tot} \lesssim 0.2$ and 2×10^6 GeV for $c_{\rm tot} = 36$. A pre-decay asymmetry is therefore diluted by 10^{-9}–10^{-26}, and lepton-violating decays are converted with $\eta_L \lesssim 10^{-8}$ except in the upper part of the LVS window. What survives is baryon-violating decay, and the required baryon number per decay, $\epsilon_{\rm req} \sim 10^{-4}$–$10^{-3}$, is the same in all four constructions. It is a property of the entropy wall, not of the stabilisation scheme. LVS differs in whether the decay chain that would supply ϵ exists: the gaugino-pair channel used in [22] is open for the first three and closed for LVS.

Channel	Condition	$\mathcal{V}$ range
Pre-decay EW baryogenesis	$T_R \gtrsim 6$–38 GeV	$\lesssim (1.7\text{–}3.8)\times10^7$
Axion velocity, bulk/fibre	$F \lesssim 4\times10^{10}$ GeV	excluded ($F \simeq 9\times10^{15}$)
Axion velocity, blow-up	$\Gamma_\rho \gtrsim H_*$	excluded (fast decay)
Decay leptogenesis	$T_R \gtrsim 100$–300 GeV	$\lesssim 1.1\times10^7$
B-violating decays	$\epsilon\,\eta_B\kappa_Y T_R/m \simeq Y_B^{\rm obs}$	$\lesssim 3.1\times10^{12}\epsilon^{4/3}$
B-violating decays, sequestered	$\epsilon \simeq (1\text{–}6)\times10^{-4}$; gaugino channel closed, higgsino channel suppressed or requires tuned μ	1.5×10^7–1.5×10^8
Affleck–Dine, baryonic	see Sec. 7	$\lesssim 6\times10^5\,\zeta$; $\lesssim 10^4$ in the effective theory

Table 5: Summary of constraints on LVS, assuming a single modulus-dominated era and no thermal inflation.

7 Affleck–Dine baryogenesis

7.1 Setup

For a flat direction lifted by $W = \lambda\phi^n/(nM_*^{n-3})$,

$$V = (m_{\rm AD}^2 - cH^2)|\phi|^2 + \frac{\lambda^2|\phi|^{2n-2}}{M_*^{2n-6}} + \Big(\frac{(a\,m_{\rm AD}e^{i\alpha} + a_H H e^{i\alpha_H})\lambda\phi^n}{nM_*^{n-3}} + \text{h.c.}\Big). \tag{21}$$

For $H \gg m_{\rm AD}$ the field tracks $|\phi_0|^2 = \left(\sqrt{c/(n-1)}\, H M_*^{n-3}/\lambda\right)^{2/(n-2)}$ and begins to oscillate at $H = m_{\rm AD}$, leaving a charge density $n_B = \epsilon\, m_{\rm AD}|\phi_0|^2$. The relative phase $\delta = \alpha - \alpha_H$ is the CP-violating input. The *udd* direction has $n = 6$; LH_u has $n = 4$. The operator scale M_* is not fixed by the four-dimensional analysis; we write $M_* = \zeta M_{\rm Pl}$. Sec. 7.5 shows that the result depends sharply on ζ.

7.2 Yield

We consider $m_{\rm AD} > m_{\rm mod}$, the generic LVS ordering ($m_{\rm AD} \sim m_{3/2}$, $m_{\rm mod}/m_{3/2} \sim \mathcal{V}^{-1/2}$), with inflaton oscillations until $H \simeq m_{\rm AD}$ and reheating at $T_{\rm RH} \le T_{\rm AD}$, $3m_{\rm AD}^2 M_{\rm Pl}^2 = \frac{\pi^2}{30} g_* T_{\rm AD}^4$. The maximal yield is $Y_{\rm AD} = \epsilon|\phi_0|^2 T_{\rm AD}/4 m_{\rm AD} M_{\rm Pl}^2$; for lower $T_{\rm RH}$ it is reduced by $T_{\rm RH}/T_{\rm AD}$, so what follows are upper bounds. The modulus later oscillates at $H = m_{\rm mod}$ with energy fraction $r_i = \Delta^2/6M_{\rm Pl}^2$ and produces the dilution $\mathcal{D} = \frac{4}{3}\kappa_Y (T_R/T_i)(1-r_i)/r_i$, $T_i \propto \sqrt{m_{\rm mod} M_{\rm Pl}}$. For $n = 6$ the dependence on $m_{\rm AD}$ cancels, and

$$Y_B \simeq 0.13\, \eta\, \epsilon\, \zeta^{3/2} \sqrt{c_\Gamma}\, \frac{m_{\rm mod}}{M_{\rm Pl}} \qquad (r_i = \tfrac{1}{6},\ c = \lambda = 1,\ g_* = 106.75). \tag{22}$$

The coefficient is 0.24 for $g_R = 10.75$; the dependence on c and λ is weak ($c^{1/4}\lambda^{-1/2}$), while $Y_B \propto 1/r_i$, so $\Delta = M_{\rm Pl}$ is the conservative choice. Because $T_R \gg T_*$ in the relevant range, $\eta = 28/79$ for both baryonic and leptonic directions. For $\zeta = 1$, $\epsilon = 0.51$, $Y_B = Y_B^{\rm obs}$ requires $m_{\rm mod} \gtrsim 8.7 \times 10^9$ GeV.

7.3 Charge efficiency

We integrate the field equations of (21) in a matter-dominated background ($H = 2/3t$) in units $\phi = M_{\rm Pl} f$, $\tau = m_{\rm AD} t$, $h = H/m_{\rm AD}$. For $h \gg 1$ the equations have a scale-free attractor $f = R\, t^{-1/(n-2)} e^{i\theta_{\rm min}}$ with $n\theta_{\rm min} + \arg a_{\rm tot} = \pi$; starting on it with the exact velocity removes transients that otherwise change ϵ by a factor two. The comoving charge is normalised to $a^3 m_{\rm AD}|\phi_0|^2$ at $H = m_{\rm AD}$, which defines ϵ. For the benchmark $a = a_H = c = \lambda = 1$, $\delta = \pi/2$ we find $|\epsilon| = 0.51$ ($n = 6$) and 0.69 ($n = 4$). A scan of 162 points ($c, a, a_H \in \{0.3, 1, 3\}$, six values of δ) gives median $|\epsilon| = 0.16$, 90th percentile 0.55, maximum 1.2. The benchmark has maximal CP phase; the median is the more representative value.

7.4 Results for $M_* = M_{\rm Pl}$

For $m_{\rm AD} = 0.1\, m_{3/2}$, $r_i = \frac{1}{6}$, $c_\Gamma = 1$, the maximal volume for which $Y_B^{\rm obs}$ is reached is, for $\epsilon = 1.2$, 0.51, 0.16:

	$\epsilon = 1.2$	0.51	0.16
udd, Swiss-cheese	1.1×10^6	5.9×10^5	2.8×10^5
udd, K3-fibred	2.5×10^5	1.5×10^5	7.3×10^4
LH_u, Swiss-cheese	1.9×10^4	1.2×10^4	6.9×10^3

At $\mathcal{V} = 5.9 \times 10^5$, $m_{3/2} = 4.1 \times 10^{12}$ GeV, $m_{\rm mod} = 8.7 \times 10^9$ GeV and $T_R = 7.9 \times 10^4$ GeV. The leptonic direction fails because $|\phi_0|^2$ is smaller than for $n = 6$ by $0.87\sqrt{m_{\rm AD}/M_{\rm Pl}} \sim 10^{-4}$–$10^{-3}$. If the soft terms are sequestered below the modulus mass ($m_{\rm AD} < m_{\rm mod}$), the yield is $\propto T_R/\sqrt{m_{\rm AD} M_{\rm Pl}}$ and the window extends to $\mathcal{V} \sim 10^7$ for $m_{\rm AD}/m_{\rm mod} \sim 10^{-6}$.

7.5 Consistency of the effective theory

In LVS the string and Kaluza–Klein scales are $M_s \simeq g_s^{1/4} M_{\rm Pl}/\sqrt{4\pi\mathcal{V}}$ and $M_{\rm KK} \sim M_{\rm Pl}\mathcal{V}^{-2/3}$; at $\mathcal{V} = 5.9\times10^5$, $M_s \approx 7\times10^{14}$ GeV and $M_{\rm KK} \approx 3\times10^{14}$ GeV (up to $O(1)$ convention-dependent factors). With $m_{\rm AD} = 0.1\, m_{3/2}$,

$$\frac{T_{\rm AD}}{M_s} \simeq 0.8, \qquad \frac{|\phi_0|}{M_s} \simeq 2.2\,\mathcal{V}^{1/4}\,\zeta^{3/4} \qquad (n=6). \tag{23}$$

The reheating temperature at which Eq. (22) is saturated lies at the string scale for every $\mathcal{V}$, and for $\zeta = 1$ the flat direction is displaced to $\approx 60\, M_s$ at $\mathcal{V} = 5.9\times10^5$. The four-dimensional analysis is then outside its range of validity. The charge efficiency ϵ is independent of ζ, since ζ can be absorbed into λ in Eq. (21); ζ enters only through $|\phi_0|^2 \propto \zeta^{3/2}$. Evaluating the full yield, Eq. (22) without the approximations of the coefficient, in the explicit vacua (and in the K3 vacua with the fibre modulus) gives Table 6. The scan reproduces the values for $\zeta = 1$ above to within 6%, and $\mathcal{V}_{\rm max} \propto \zeta$ to within 15%. Three prescriptions for M_* are of interest: $\zeta = 1$; M_* chosen so that $|\phi_0| = M_s$; and $M_* = M_s$.

	Swiss-cheese			K3-fibred		
ϵ	1.2	0.51	0.16	1.2	0.51	0.16
$\zeta = 1$	1.0×10^6	6.0×10^5	2.8×10^5	2.4×10^5	1.5×10^5	7.3×10^4
$\zeta = 0.1$	1.1×10^5	6.4×10^4	3.0×10^4	3.1×10^4	1.8×10^4	9.2×10^3
$\lvert\phi_0\rvert = M_s$	1.6×10^4	1.1×10^4	6.0×10^3	6.7×10^3	4.5×10^3	2.7×10^3
$M_* = M_s$	4.0×10^3	2.8×10^3	–	2.0×10^3	1.4×10^3	–

Table 6: Maximal volume for baryonic Affleck–Dine baryogenesis ($n = 6$, $m_{\rm AD} = 0.1\, m_{3/2}$, $c_\Gamma = 1$, $r_i = \frac{1}{6}$) as a function of the operator scale and the charge efficiency. A dash indicates no solution for $\mathcal{V} \geq 2.5\times10^3$ (Swiss-cheese) or 10^3 (K3).

Within the effective theory the maximal volume is $\mathcal{V} \sim 10^4$ (Swiss-cheese) and 5×10^3 (K3-fibred), and for $M_* = M_s$ it is $\sim 3\times10^3$ and 1.4×10^3, where the large-volume expansion is no longer under control.

The allowed region is shown in Fig. 5. Several further effects bear on the analysis: (i) the blow-up modulus and its axion have $m \simeq (17\text{–}78) m_{3/2} \gg m_{\rm AD}$, so they begin to oscillate before $H = m_{\rm AD}$ and decay at $H \sim 10^4$ GeV, long after, and may dominate the energy density across the epoch in which the flat direction is displaced; Case I assumes inflaton-driven expansion. (ii) The condensate can fragment into Q-balls; this changes the dynamics but not the total charge. (iii) With $m_{\rm AD} \gtrsim 10^9$ GeV the low-energy theory is not supersymmetric and dark matter must be supplied separately. (iv) The sequestered case is more favourable, since $|\phi_0| \propto m_{\rm AD}^{1/4}$, but the analysis of Hubble-induced terms during modulus domination is beyond the scope of this paper.

The conclusion is that baryonic Affleck–Dine baryogenesis with gravity-mediated soft terms does not give a controlled window in LVS at large volume; it is possible at $\mathcal{V} \lesssim 10^4$ (K3: 5×10^3) or with sequestered soft terms. Affleck–Dine baryogenesis in LVS was studied in [20], who find that it can generate the asymmetry (with or without an enhanced inflaton coupling, and with Q-ball formation); we have not reproduced their treatment of the dilution and of Q-balls, and the comparison with the effective-theory analysis above remains to be made. The present analysis concerns the case in which the modulus itself dilutes the asymmetry, as in [19, 24].

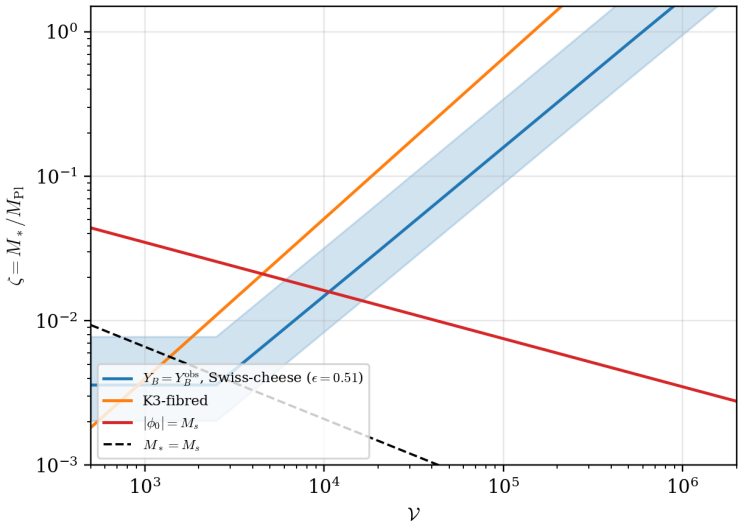


Figure 5: Baryonic Affleck–Dine baryogenesis in the plane of volume and operator scale $\zeta = M_*/M_{\rm Pl}$. The observed asymmetry is reached above the blue line (Swiss-cheese, $\epsilon = 0.51$; band: $\epsilon = 0.16$–1.2) or the orange line (K3-fibred); the effective theory is valid below the red line, $|\phi_0| = M_s$. The dashed line is $M_* = M_s$.

8 Discussion

Within the assumptions stated (a single modulus-dominated era, no thermal inflation), electroweak-scale baryogenesis before the decay, axion-velocity mechanisms with bulk, fibre or blow-up axions, and leptogenesis from modulus decay are excluded in LVS at the volumes that give TeV-scale supersymmetry. What survives is baryon-violating modulus decay. The requirement, 10^{-4}–10^{-3} baryons per decay, was established for the IIB, heterotic and G_2 constructions in [22]; it follows from the entropy wall and holds in LVS as well (Table 4). The gaugino channel that supplies it in those constructions is closed in LVS. The Higgs channel is suppressed, $\mathrm{Br}_{\tilde{h}\tilde{h}} \lesssim 10^{-4}$, when the cancellation in μ is structural, and reaches $\mathrm{Br} \simeq 1$ only if μ is tuned; the gluino-loop asymmetry then reaches the required size only with a bino admixture or light, strongly mixed stops. Affleck–Dine baryogenesis has a controlled regime only at small volume or with sequestered soft terms. The sphaleron results of Sec. 3 apply to any modulus-dominated cosmology.

Limitations. (1) The lattice parametrisation is discontinuous at T_c and c is calibrated; efficiencies are uncertain at $O(1)$ through c, though the exponent in $\eta_L \propto T_R^4$ is robust. (2) $g_*(T)$ is approximated near the QCD crossover. (3) The Swiss-cheese vacua are AdS and use $W_0 = A = 1$; c_Γ is a parameter, and varying it over $[0.3, 8]$ changes $\mathcal{V}_{\max}$ by a factor $\lesssim 2$. (4) The K3 loop potential is conjectural [9]; variations in the loop data change $\mathcal{V}_{\max}$ by a factor $\lesssim 4$. (5) All bounds assume the lightest modulus dominates (Sec. 2) and, for $T_R \lesssim 1$ GeV, a specific pre-oscillation history. (6) The kinetic bound assumes Chern–Simons-driven baryogenesis; non-sphaleron axion mechanisms are not constrained. (7) ϵ for baryon-violating decays and the string-theoretic origin of the A-terms are not derived. (8) The Higgs-sector couplings depend on the matter Kähler metric $\hat{Z}$, which we take $\propto \mathcal{V}^{-2/3}$; the Higgs soft masses for this choice are far above the TeV scale.

Data and code. All numerical results are generated by the code at `https://github.com/vishal1601-2005/Entropy-walls-LVS` (MIT licence), which contains the intermediate data and the LaTeX source. The inputs are the lattice rate [18], the K3 parameter sets [9] and $Y_B^{\rm obs}$.

A Dilution coefficient

Let the sudden-decay entropy be $S_{\rm fin} = \frac{2\pi^2}{45} g_R T_R^3 a_d^3$, with $\rho_\phi(a_d) = 3M_{\rm Pl}^2\Gamma^2$. Since $\rho_\phi \propto H^2$ and $\rho_\phi \propto a^{-3}$, $a_d^3/a_{\rm gen}^3 = H_{\rm gen}^2/\Gamma^2$, with $H_{\rm gen}$ from Eq. (3). Hence

$$\mathcal{D} = \frac{g_*}{g_R}\Big(\frac{T_{\rm gen}}{T_R}\Big)^3\Big(\frac{\Gamma}{H_{\rm gen}}\Big)^2, \qquad \frac{\Gamma}{H_{\rm gen}} = \frac{b}{\kappa}\Big(\frac{T_R}{T_{\rm gen}}\Big)^4, \qquad \frac{b}{\kappa} = \frac{36\,b^2}{\pi^2 g_*} = 0.4\,\frac{g_R}{g_*}, \tag{24}$$

with $b = \sqrt{\pi^2 g_R/90}$. Therefore $\mathcal{D} = 0.16\,(g_R/g_*)(T_R/T_{\rm gen})^5$, which is Eq. (4). The same ratio gives $\Gamma/H(T) = \frac{2g_R}{5g_*}(T_R/T)^4$ used in Sec. 3.3.

B Susceptibility

In units of $T^2/6$ per species, with flavour-universal potentials, hypercharge neutrality reads $n_g[\mu_Q + 2\mu_u - \mu_d - \mu_L - \mu_e] + 2\mu_H = 0$ (the Higgs term carries the bosonic factor). With $\mu_u = \mu_Q + \mu_H$, etc., this gives $\mu_H = \frac{3}{7}(\mu_L - \mu_Q)$. Then $B = 12\mu_Q$, $B - L = 12\mu_Q - 9\mu_L + 3\mu_H$, and $\mu_{\rm CS} = 9\mu_Q + 3\mu_L$. For $B - L = 0$, $\mu_L = \frac{25}{18}\mu_Q$ and $\mu_Q = \frac{6}{79}\mu_{\rm CS}$, so $n_B = \frac{12}{79}T^2\mu_{\rm CS}$. For $\mu_{\rm CS} = 0$, $\mu_L = -3\mu_Q$ and $B/(B-L) = 28/79$.

C Kähler metric of the blow-up modulus

For $K = -2\ln\mathcal{V}$ with $\mathcal{V} = \gamma(\tau_b^{3/2} - \tau_s^{3/2})$ and $\tau = \frac{1}{2}(T + \bar T)$, $K_{s\bar s} = \frac{1}{4}\partial_{\tau_s}^2 K$. For $\tau_s^{3/2} \ll \tau_b^{3/2}$, $K = -2\ln(\gamma\tau_b^{3/2}) + 2\tau_s^{3/2}/\tau_b^{3/2} + \ldots$, hence $K_{s\bar s} \simeq 3\gamma/8\sqrt{\tau_s}\mathcal{V}$. We checked this against the exact second derivative at $\mathcal{V} = 1.5\times10^{14}$, $\tau_s = 25.1$ (four digits).

D Explicit amplitude check of the CP asymmetry

With the stops integrated out, the interactions are

$$\mathcal{L}_{\rm eff} = \sum_{\alpha\beta} D_{\alpha\beta}\,J_\alpha^* J_\beta, \qquad J_\alpha = \sum_{i=1,2}\Big[A_{i\alpha}(\lambda_i\psi_Q) + B_{i\alpha}(\lambda_i^\dagger\psi_{U^c}^\dagger)\Big] + \delta_{\alpha R}\,\rho\,(\psi_{b^c}\psi_{s^c}), \tag{25}$$

with $\alpha \in \{R, L\}$, $D = (M_{\tilde t}^2)^{-1}$, and $\lambda_{1,2}$ the Majorana fermions. For the gluino $A_L = -\sqrt{2}g_3$, $B_R = \sqrt{2}g_3$; for the neutral higgsino $A_R = B_L = -y_t/\sqrt{2}$; for a bino admixture $B_R = \frac{2\sqrt{2}}{3}g_1 a_B$ and $A_L = -\frac{\sqrt{2}}{6}g_1 a_B$. The tree amplitudes for $\lambda_1 \to tsb$ and $\bar t\bar s\bar b$, the three-body amplitude for $\lambda_1 \to t\lambda_2\bar t$ (two diagrams, since λ_2 is Majorana) and the amplitude for $\lambda_2\bar t \to bs$ are evaluated with explicit two-component spinors. The absorptive part follows from unitarity,

$$\mathcal{T}_{fi}^{\rm abs} = \tfrac{i}{2}\sum_n \int d\Phi_n\,\mathcal{T}_{fn}\mathcal{T}_{ni}, \qquad n = (\lambda_2, t, \bar t), \tag{26}$$

with the top a spectator, and the asymmetry is $\Gamma(f) - \Gamma(\bar f)$ integrated over the three-body phase space. The relative sign of the f and $\bar f$ cuts is a fermion-ordering sign, which we fix by CP invariance:

at zero Majorana-mass phases the raw $\bar{f}$ contribution equals exactly minus the f contribution for every coupling set tested (pure right-handed, mixed, higgsino, bino–higgsino), whereas CP requires them to be equal.

The implementation reproduces the tree rate (39) of [22] for all stop mixings and couplings tested, and the cross-chirality terms of (40). For right-handed couplings it reproduces the f_2 term of (41), with the dependence on $x = m^2_{\tilde{\lambda}_2}/m^2_{\tilde{\lambda}_1}$ confirmed at $x = 0.09$, 0.25, 0.49 and the dependence $\sin\phi_{12}$ on the phase. It differs from (41) in two respects. First, the left-handed term $\kappa^2_{1L}\kappa^2_{2L}c_t^2 s_t^4$ enters with the opposite sign to the right-handed one. Second, there is no f_1 term. Both are consistent with rephasing invariance: a term without a gaugino mass insertion cannot depend on the phase of the gaugino mass, and the right- and left-handed mass-insertion terms carry opposite phases. As a check of the phase handling, the asymmetry depends on the phases only through $\phi_2 - \phi_1$ (four different pairs with the same difference agree to six digits, and the sign reverses with the difference), as rephasing invariance requires. The same-chirality three-body rate differs from (40) by 3% at $x = 0.25$, and the cross-chirality terms of (40) agree after a rescaling of the couplings by $\sqrt{2}$, which we do not understand; neither affects the right-handed benchmark (47) or the large-λ'' limit. The conclusions of Sec. 6.3 are stated for both signs of the left-handed term.

We have checked the relative sign analytically. In the crossed diagram that carries the gaugino mass insertion, the coupling products of the right- and left-handed chains are

$$C_R \propto y_L^2\kappa^2_{2R}\, e^{-i\Phi}, \qquad C_L \propto y_R^2\kappa^2_{2L}\, e^{+i\Phi}, \qquad \Phi = \phi_2 - \phi_1 + 2\arg D_{LR}, \tag{27}$$

with the common prefactor $-|\rho|^2 D_{RR}|D_{LR}|^2$, where $y_{R,L}$ are the couplings of the decaying fermion to the structures $(\tilde{h}\psi_Q)\tilde{t}^*_R$ and $(\tilde{h}\psi_{U^c})\tilde{t}_L$. They are complex conjugates up to the magnitudes of the couplings. The spin-summed kinematic factors are equal, $S_R = S_L = 2m_{\lambda_1} m_{\lambda_2}\, p_t\cdot l_{\bar{t}}$, as follows from the identities $\sum x_\alpha y^\beta = m\,\delta_\alpha{}^\beta$ and $\sum x_\alpha x^\dagger_{\dot\beta} = \sum y_\alpha y^\dagger_{\dot\beta} = (p\cdot\sigma)_{\alpha\dot\beta}$, and from explicit spinors. Hence the CP-odd parts combine to $(y_L^2\kappa^2_{2R} - y_R^2\kappa^2_{2L})\sin\Phi$, which vanishes for a pure higgsino ($y_L = y_R$) and the gluino ($|\kappa_{2R}| = |\kappa_{2L}|$). A chirality-mixed chain, the origin of the f_1 term, carries the coupling product $\kappa_{1R}\kappa_{1L}\kappa_{2R}\kappa_{2L}$, which is independent of ϕ_1 and ϕ_2. For a mixed neutralino there are several invariants, and the asymmetry of the benchmark of Sec. 6.3 is within a factor 1.9 of its maximum over ϕ_{12} and $\arg X_t$. These checks use the same operator expansion as the numerical calculation; a check with an independent diagrammatic tool is desirable. The unitarity sum over the final states $\{f, \bar{f}\}$ is trivially satisfied for a Majorana parent.

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